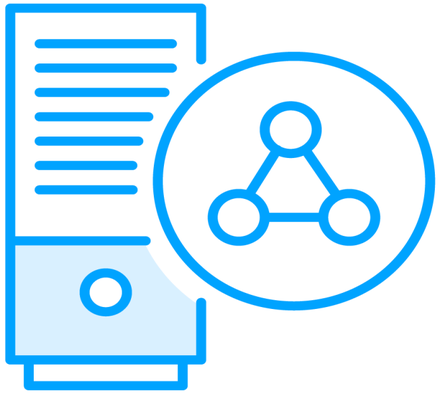
Federated Learning and Privacy-preserving RAGs

Luca Berton

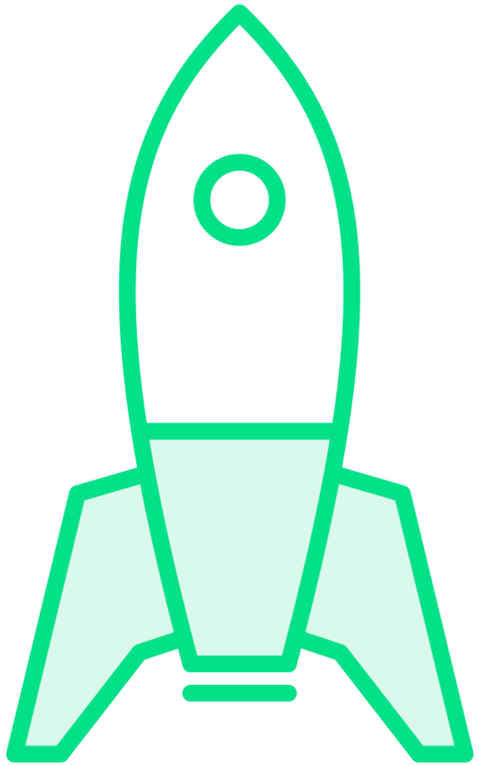
AI Expert

@BertonLuca | www.lucaberton.it



Federated Learning

Privacy-preserving



Train RAG models in a decentralized way

Secure and privacy-preserving retrieval techniques

Implement federated learning algorithms and privacy-preserving mechanisms

Design and deploy RAG systems with privacy regulations

Experience with AI and ML systems

Retrieval-Augmented Generation

Federated Learning and Privacy-preserving RAGs

Federated Learning for Retrieval

Luca Berton

AI Expert

@BertonLuca | www.lucaberton.it

Warning – The principles and examples in this course will never be complete!

Federated Learning for Retrieval

Share
only model updates
not the raw data

Advantages of Federated Learning

Data Privacy

Reduced Bandwidth Usage

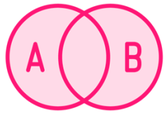
Model Generalization

Regulatory Compliance

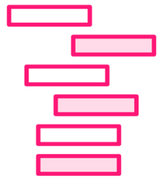
Scalability

Federated Learning for Retrieval-Augmented Generation (RAG)

Enhancing Generative AI



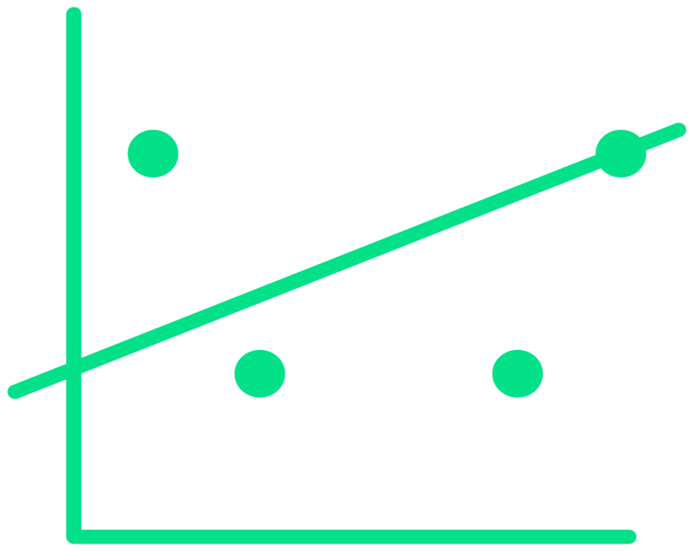
Combining Techniques in AI



Enhanced Content Generation

Addressing Challenges with Advanced Techniques

FedAvg

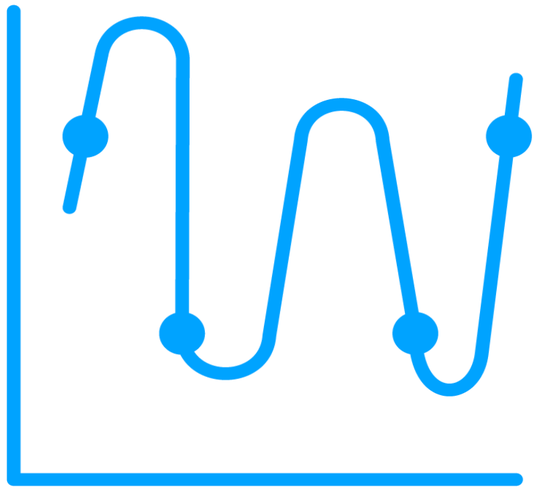


Federated Averaging

- Aggregates model updates
- Receive only the results
- Update in a global model

Best for privacy-focused training across multiple decentralized devices.

FedSGD



Federated Stochastic Gradient Descent

- Traditional stochastic gradient descent (SGD)
- Receive the gradients from clients
- Update in a global model

Ideal for continuous learning from diverse datasets while maintaining decentralization

Federated Transfer Learning

Federated Transfer Learning

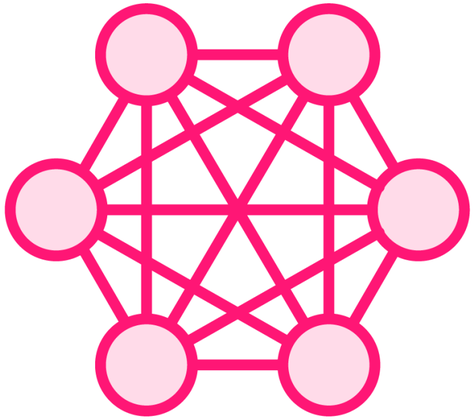
- Learn from decentralized sources

Excels in adapting models to varying data distributions and feature spaces across different clients.

Use Case: Healthcare

Patient data locally

Add new data

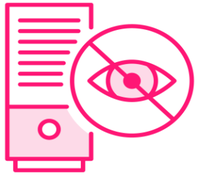


Enhancing the model accuracy

Privacy-Preserving Techniques in Federated Learning

Data Privacy in AI

Decentralized Training

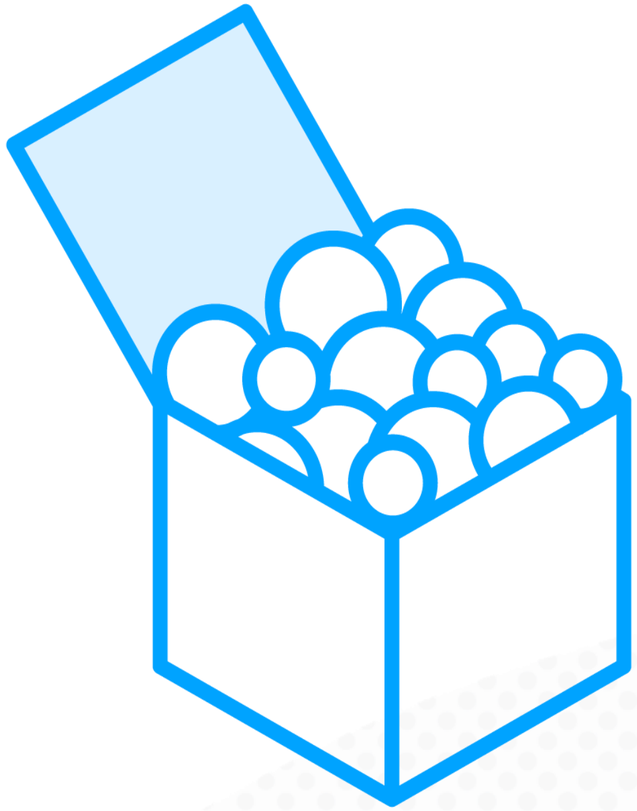


Privacy Enhancement

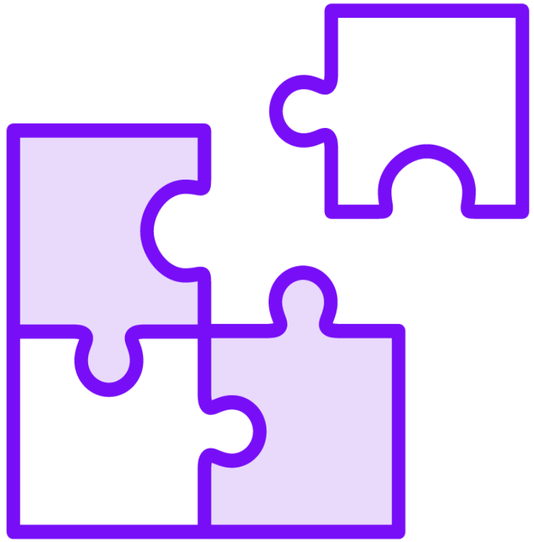
Privacy-Preserving Techniques

Homomorphic Encryption

Secure Calculations

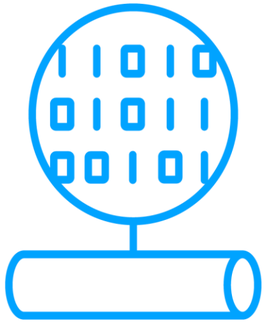


Differential Privacy



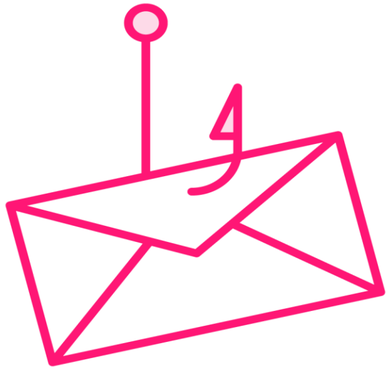
Secure Multi-Party Computation (SMPC)

Use Case: Finance



Encrypted transactions

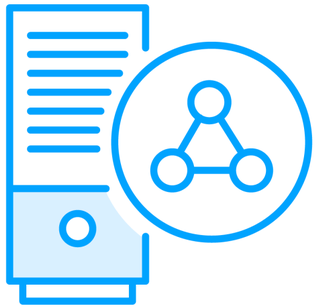
Protect personal information



Share fraud detections

Implementing Privacy-Preserving RAG in Federated Learning

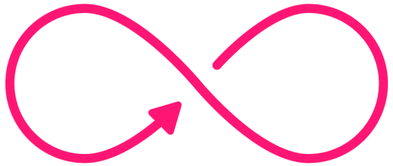
Implementing Privacy-Preserving RAG in Federated Learning



Decentralized Model Training

Privacy-Preserving Techniques

RAG Model Enhancement

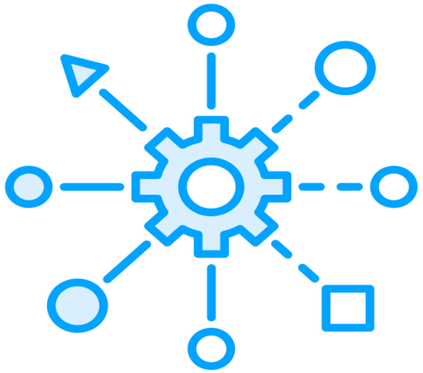


Continuous Improvement

Use Case: Retail

Customer data

Purchase trends



Product recommendations

Summary

Use cases:

- Healthcare
- Finance
- Retail

Federated learning leverages techniques

- Federated Averaging (FedAvg)
- Federated Stochastic Gradient Descent (FedSGD)
- Federated Transfer Learning

Privacy-preserving techniques

- Homomorphic Encryption
- Differential Privacy
- Secure Multi-Party Computation (SMPC)